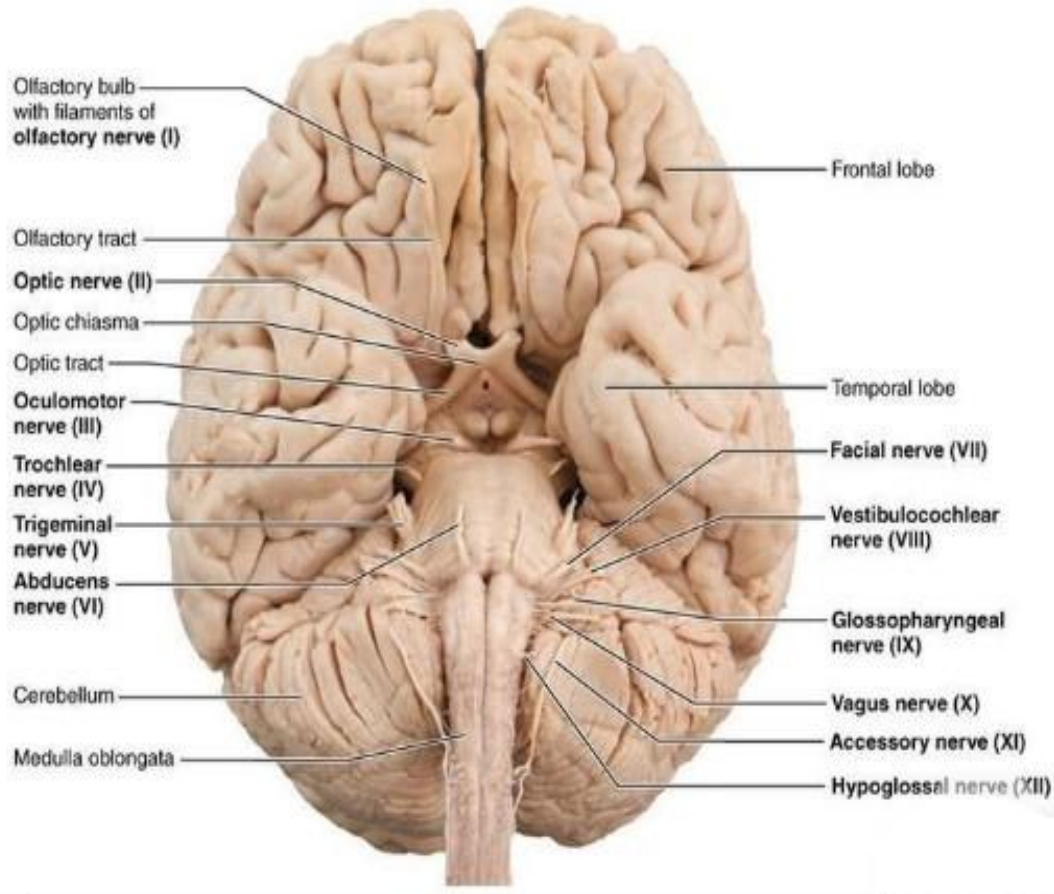


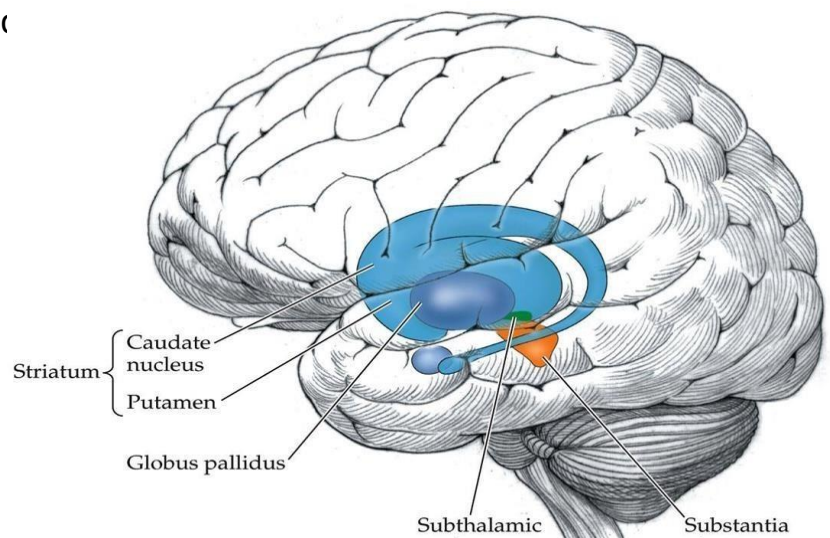
- 1- **Anterior cerebral artery** supplies anterior part of anterior limb.
- 2- **Middle cerebral artery:** Supplies posterior part of anterior limb, genu & lenticulothalamic part.
- 3- **Anterior choroidal artery:** Supplies sublenticular, retro lenticular parts and ventral half of posterior limb.



Lecture (15): cerebrum II

Basal Nuclei (Basal Ganglia)

- The **basal ganglia (BG)** consist of **subcortical interconnected nuclei** which are deeply situated inside the white matter of the cerebral hemispheres.
- They are involved in the **initiation** and
- **Components of BG:**
 - **Caudate nucleus, putamen, globus pallidus, amygdaloid nucleus & nucleus accumbens** (in the telencephalon).



- **Subthalamic nucleus** (in the diencephalon).
- **Substantia nigra** (in the mesencephalon).
- **The caudate nucleus and the putamen** are two nuclei that are interconnected and form the **striatum**.
- **Nucleus Accumbens** and part of **caudate** form **Ventral striatum**.

▣ **Caudate Nucleus:**

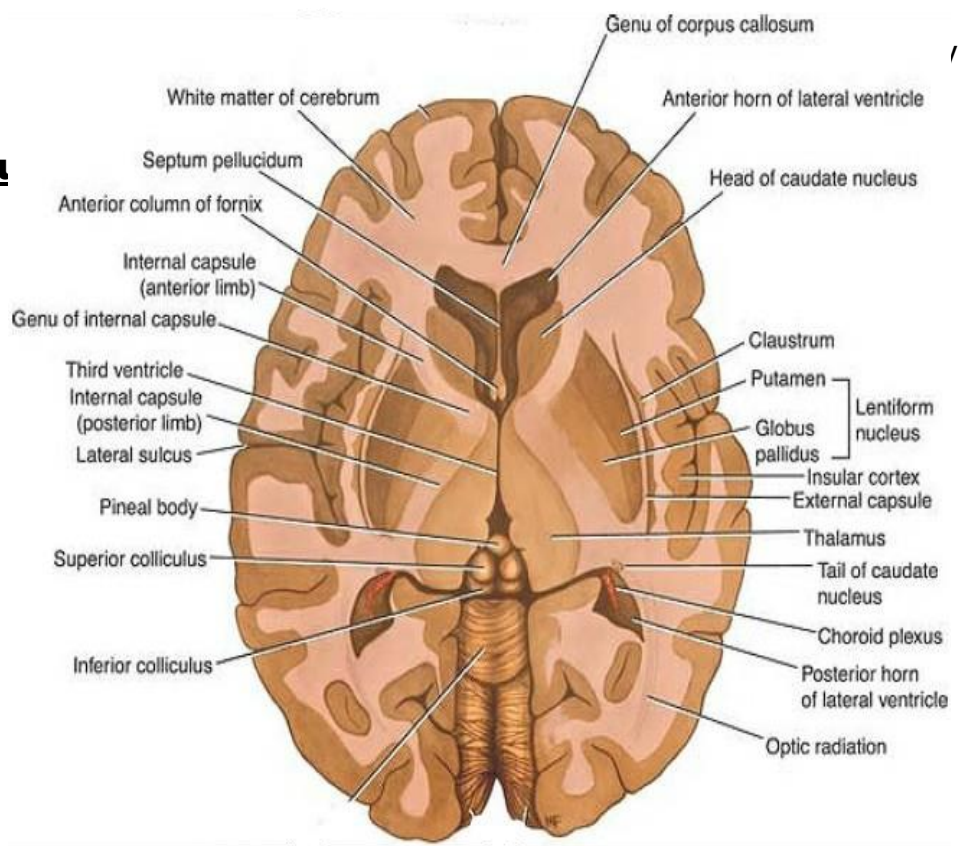
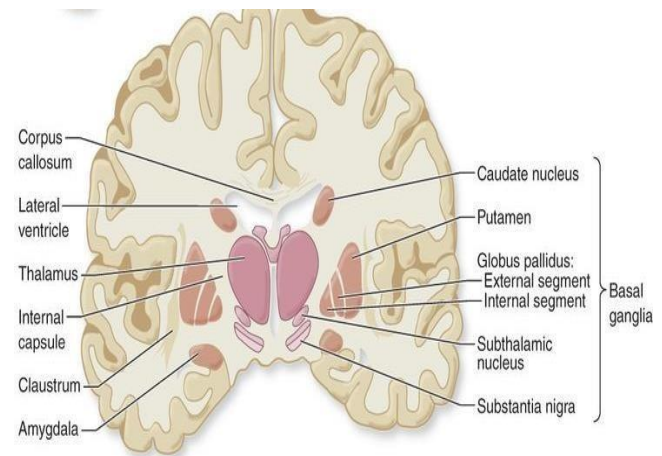
- The caudate nucleus is a large **C-shaped mass of grey matter** that is closely related to the lateral ventricle.
- The lateral surface of the nucleus is related to the internal capsule, which separates it from the lentiform nucleus.
- It is divided into a **head**, a **body**, and a **tail**.
- **Head** lies in the lateral wall of the anterior horn of the lateral ventricle.

It is separated from lentiform nucleus by anterior limb of internal capsule.

- **Body** lies in the floor of the body of the lateral ventricle.
- **Tail** lies in the roof of the amygdaloid nucleus.

▣ **Lentiform nucleus**

- Lentiform nucleus is a **wedge-shaped mass of grey matter** whose broad convex base is directed laterally, and apex is directed medially.
- It is related medially to the **internal capsule**, which separates it from the caudate

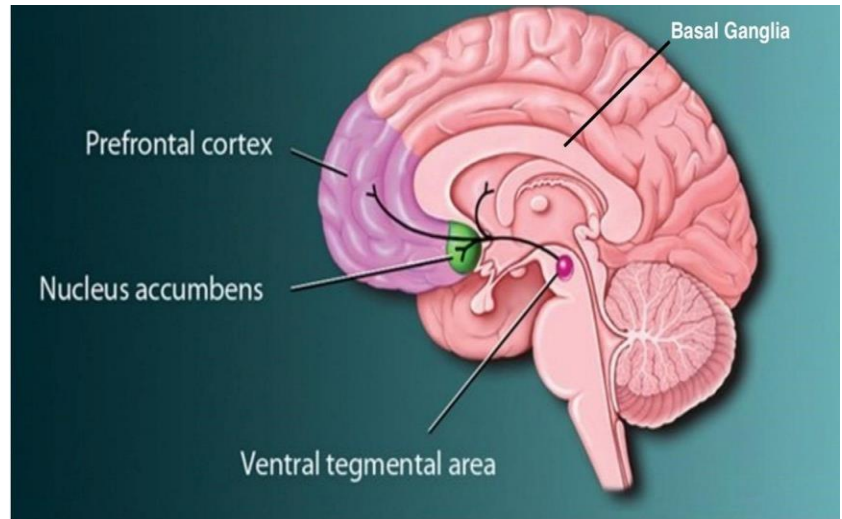


nucleus and the thalamus.

- It is related laterally to the **external capsule** which separates it from the **claustrum**.
- It is divided into:
 - The **putamen** is a larger, darker lateral portion.
 - The **globus pallidus** is an inner lighter portion.
- **Globus pallidus (GP)** is divided into: **GP interna (medial) & GP externa (lateral)**.

- **Nucleus Accumbens:**

- It lies in the rostral cerebral hemisphere, between putamen and caudate nucleus.
- This nucleus is a **major component** of the **ventral striatum**.
- It is a key part of the reward system due to its strong connections to the **ventral tegmental area** in the midbrain (motivation, reward and pleasure).



- It connects the limbic system to the motor system.
- **Substantia nigra (SN) of the midbrain:**
- It consists of an outer portion (**SN pars reticulata**) and inner portion (**SN pars compacta**).
 - **Pars compacta** contain neurons that project to the striatum, are **dopaminergic**.
- Striatal dopamine excites direct circuits and inhibits indirect circuits.
- **Subthalamic nucleus of the diencephalon:**

The subthalamic nuclei are glutaminergic and excitatory to globus pallidus.

- **Amygdaloid Nucleus:** (part of limbic system) It is situated in the anterior end of inferior horn of the lateral ventricle in temporal lobe.

Connections of BG

- ▣ The **caudate nucleus** and the **putamen** form the main sites for **receiving input** to the basal nuclei.

■ The **globus pallidus** forms the major site from which **the output** leaves the basal nuclei.

■ They receive no direct input from or output to the spinal cord.

■ **Afferent Fibers:**

- **Corticostriate Fibers:** from all parts of the cerebral cortex.
- **Thalamostriate Fibers:** from the thalamus.
- **Nigrostriate Fibers:** from substantia nigra.

■ **Efferent Fibers:**

- **Striatopallidal Fibers:** to the globus pallidus.
- **Striatonigral Fibers:** to the substantia nigra.
- **Pallidofugal fibers** are complicated and can be divided into groups:
 - (1) **ansa lenticularis**, which pass to the thalamic nuclei
 - (2) **fasciculus lenticularis**, which pass to the subthalamus.
 - (3) **pallidotegmental** fibers, which terminate in tegmentum of the midbrain

Functions of BG

■ **Motivation & rewarding:** dopamine hormone is secreted from **ventral tegmental area**, **SN pars compacta** and is stored in **nucleus accumbens** which is released to **limbic system** leads to desire of movement.

■ **Control saccadic (rapid conjugate) eye movement.**

■ **Motor function:**

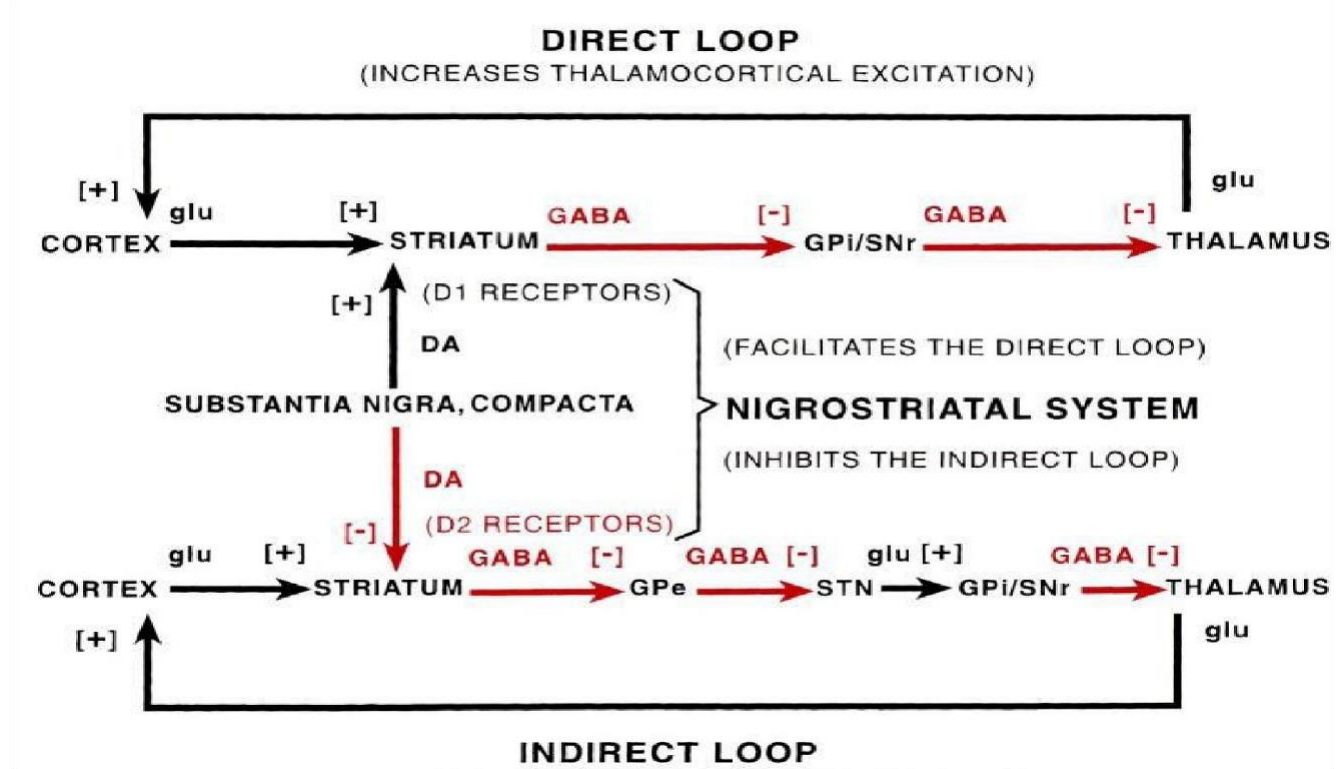
- Initiates the wanted movement through **direct circuit:** (**Cerebral cortex** projects to **striatum** which sends to **GP interna**, which projects to **AV & LV** nuclei of thalamus to excite **motor cortex** to initiate movement).
- Inhibits unwanted movement through **indirect circuit:** (**cerebral cortex** projects to **striatum** which sends to **GP externa**, which projects to, disinhibition, **subthalamic nucleus** which excites **GP interna** that sends inhibitory impulses to the **AV & LV** thalamic nuclei and **motor cortex** to inhibit movement.

Lesion to BG

- 1- **Parkinsonism:** It is associated with neuronal degeneration in substantia nigra. results in depletion of striatal dopamine. Leading to (tremors, rigidity, masked face,

shuffling gait).

- 2- **Huntington Chorea:** Lesion to striatum, involuntary, jerky, irregular movements.
- 3- **Hemiballismus:** lesion to subthalamic nucleus leads to involuntary movements.



Diencephalon

- The diencephalon is the caudal part of the forebrain.
- It is located between the cerebral hemisphere & midbrain.
- It has a single midline cavity, called the **3rd ventricle**.

- The diencephalon is subdivided into:

- 1- **Dorsal part:** Thalamus & Epithalamus.
- 2- **Ventral part:** Subthalamus & Hypothalamus.

Thalamus:

- The thalamus is a large, **egg-shaped mass of grey matter** that forms a major part of the diencephalon.
- It is situated on each side of the third ventricle.
- The anterior end of the thalamus is narrow and rounded, the posterior end is expanded to form the **pulvinar**.

- **Boundaries:**

- **Anterior end:** interventricular foramen.
- **Posterior end:** overhangs the superior colliculus.
- **Superior surface:** lateral ventricle, body of caudate nucleus, thalamostriate vein, choroid plexus of 3rd ventricle.
- **Inferior surface:** hypothalamic sulcus, subthalamus hypothalamus.
- **Medial surface:** cavity of 3rd ventricle.
- **Lateral surface:** posterior limb of internal capsule.
- Two thalamic are connected to each other by **interthalamic connection (interthalamic adhesion)**.
- Medial & lateral geniculate bodies are connected to pulvinar.

Internal structure of the thalamus:

- It is composed of grey matter, interrupted by two bands of white matter:

- **External medullary lamina** is located laterally.
- **Internal medullary lamina**, Y- shaped, It contains intralaminar nuclei. It divides thalamus into:

▣ **Anterior group:** anterior thalamic nucleus (ATN).

▣ **Medial group:** Dorso-medial nucleus (DMN).

▣ **Lateral group:** are:

- 1- Anterior ventral nucleus (AVNT).
- 2- Lateral ventral nucleus (LVNT).
- 3- Posterior medial & lateral ventral nuclei. (PMVNT – PLVNT).
- 4- Lateral dorsal nucleus.
- 5- Lateral posterior nucleus.
- 6- Pulvinar.

- **Thalamic nuclei** are divided, according to their connection with the cerebral cortex, into:
 - Specific thalamic nuclei.
 - Non-specific thalamic nuclei.

- **Specific nuclei:**

- They receive direct input from **specific ascending tracts**, and project to **specific cortical areas**.
- Subdivided into:

- **Simple relay nuclei:** project to primary sensory cortex.
- **Circuit relay nuclei:** complex relay.
- **Association nuclei:** They receive from other thalamic nuclei & project to association cortical areas.

➤ **Nonspecific nuclei:**

- project to whole the cerebral cortex for cortical arousal, attention & alertness.
- They include Reticular thalamic nuclei, Midline nuclei & Intralaminar nuclei.
- **Simple relay nuclei:**

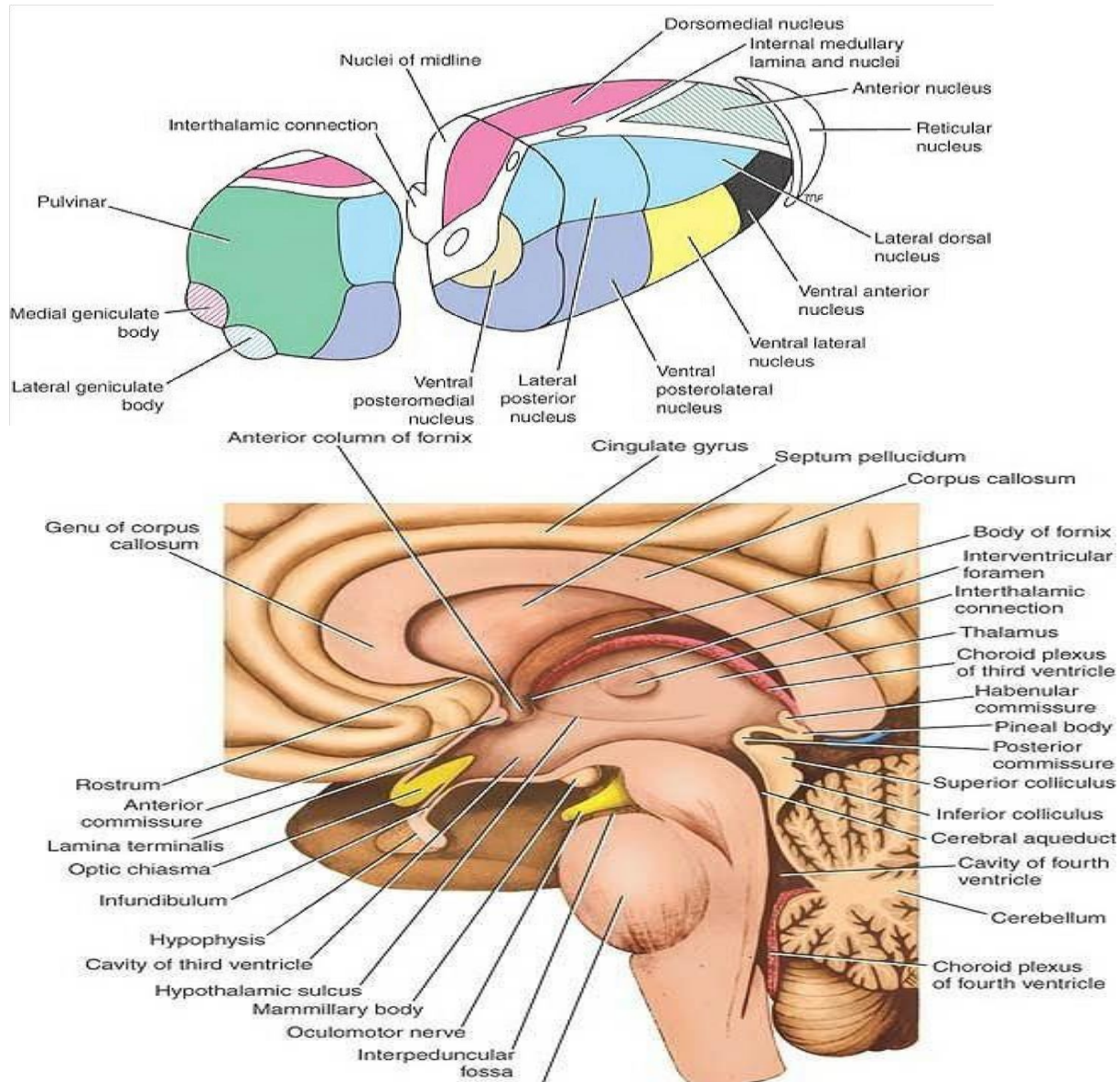
➤ Nucleus	Afferent	Efferent	Center	Function
PLVNT	Spinal lemniscus	Superior thalamic radiation.	Post-central gyrus (3,1,2)	General sensations from the body below the head.
PMVNT	Trigeminal lemniscus	Superior thalamic radiation	Post-central gyrus (3,1,2)	General sensations from face & head
LGB	Optic tract	optic radiation	Areas (17,18,19)	Vision
MGB	Lateral lemniscus	Auditory radiation	Areas (41,42)	Hearing

Circuit relay nuclei:

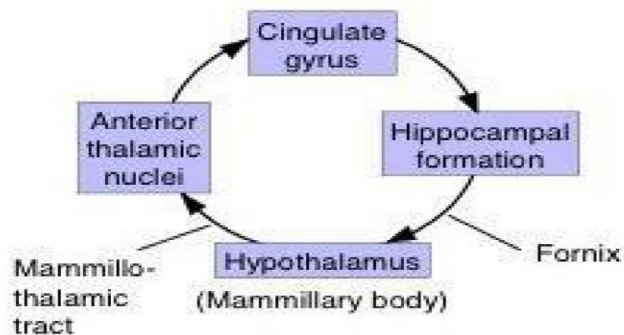
Subcortical center	Tract	Thalamic nucleus	Cortical projection
Mammillary bodies of hypothalamus	Mammillo-thalamic tract	Anterior thalamic nucleus	Cingulate gyrus
Dentate nucleus of cerebellum	Superior cerebellar peduncle	Lateral ventral nucleus	Motor cortex (areas 4,6)
Globus pallidus	Thalamic fasciculus	Anterior ventral nucleus	Motor cortex (areas 4,6)

Mainly from the **posterior cerebral artery (PCA)**.

Assisted by posterior communicating artery and anterior choroidal artery.



The Papez Circuit



Hypothalamus:

- It forms most ventral part of diencephalon.

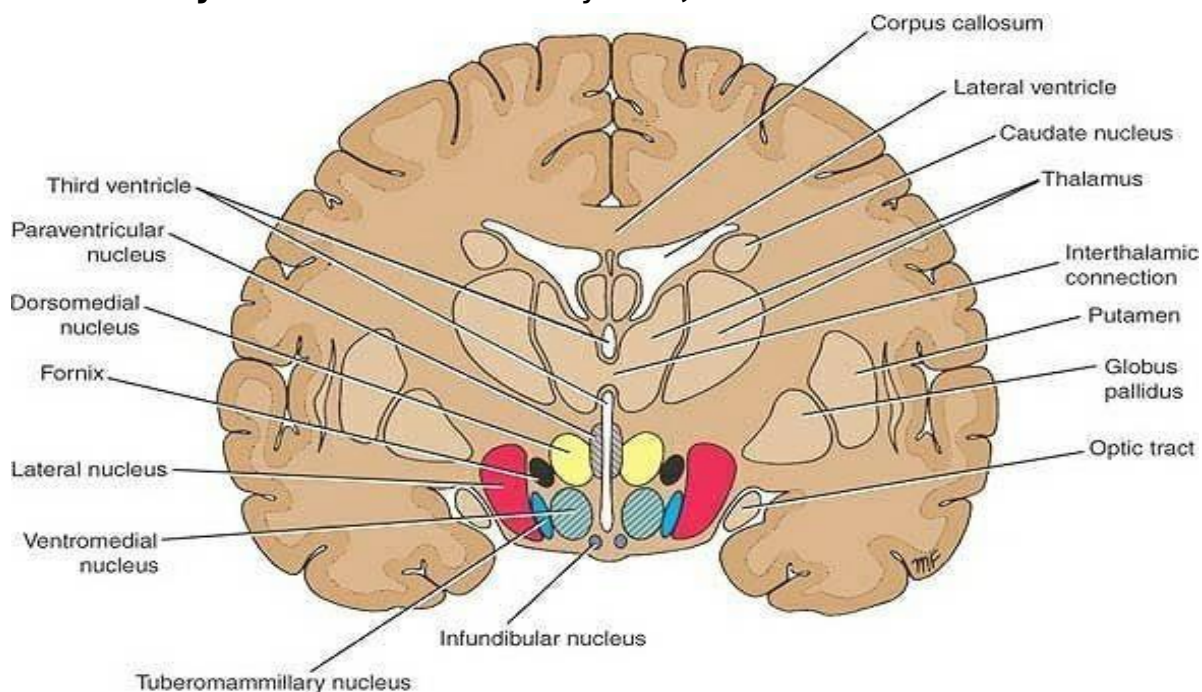
- ▣ Lies beneath the thalamus and ventromedial to the subthalamus.
- ▣ Forms the **floor** and the lower part of the **lateral wall** of the **3rd ventricle**.

▣ Boundaries of hypothalamus:

- **Anterior:** lamina terminalis.
- **Posterior & superior:** thalamus & hypothalamic sulcus.
- **Inferior:** Infundibulum, Tuber cinereum, Mammillary bodies, tegmentum of midbrain.
- **Medially:** cavity of 3rd ventricle.
- **Laterally:** subthalamus.

▣ Divisions of the hypothalamus:

- 1- **Medial zone:** related medially to 3rd ventricle, Has the following regions:
 - **Supra-optic region:** Anterior nucleus, Supra-chiasmatic nucleus, Preoptic nucleus, Paraventricular nucleus.
 - **Tuberal region:** Dorsomedial nucleus, Ventromedial nucleus, Arcuate nucleus.
 - **Mammillary region:** mammillary nuclei.
- 2- **Lateral zone:** contains lateral nucleus & supraoptic nucleus.
 - **Suprachiasmatic nucleus:** control of circadian rhythms.
 - **Supraoptic & para-ventricular nuclei:** production of vasopressin & oxytocin.
 - **Mammillary nuclei:** Part of limbic system, associated with emotions and memory.



Coronal section of diencephalon showing hypothalamic nuclei

Connection of the hypothalamus:

- The main afferent connections:

- **Fornix:** arises from hippocampus to mammillary bodies.
- **Amygdalohypothalamic fibers:** pass from amygdaloid nucleus.
- **Retino-hypothalamic fibers:** arise from retina through optic tract to supraoptic nucleus that send to pineal gland to secrete melatonin hormone.

- The main efferent connections:

- **Mammillothalamic tract:** arises from mammillary nuclei and terminates in anterior thalamic nucleus. Part of Papez circuit.

- **Hypothalamohypophyseal connections:**

- **Neural connection:** via Hypothalamohypophyseal tract, The vasopressin and oxytocin hormones are synthesized in the nerve cells of the supraoptic and paraventricular nuclei, their axons end in posterior pituitary where hormones are released.

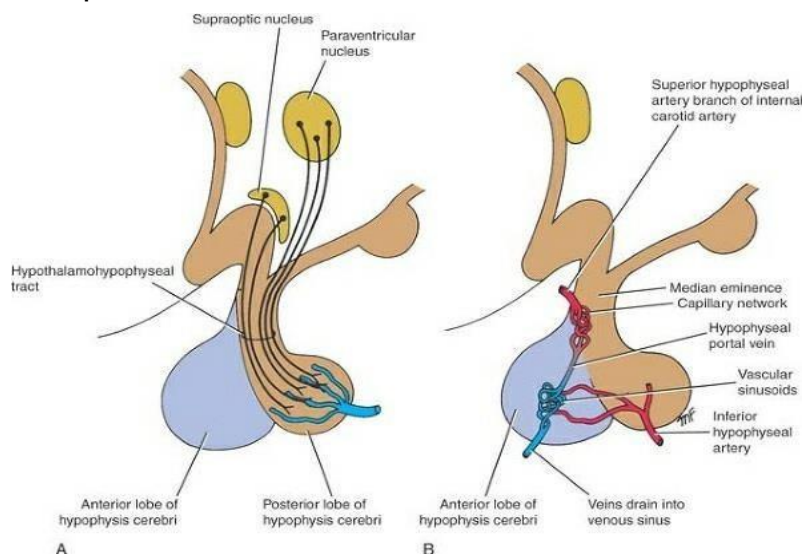
- **Vascular connection:** via

hypothalamohypophyseal portal system, which is formed by branches of superior hypophyseal artery, that enter the median eminence and divide into 1st capillary plexus. These capillaries drain into long and short descending vessels that end as 2nd capillary plexus in the anterior lobe of pituitary.

Median eminence of hypothalamus is responsible for production of releasing hormones and release inhibitory hormones, which are carried by **portal vessels** to anterior pituitary.

Limbic system:

- **Definition:** include a group of structures that lie in the medial surface of cerebral hemisphere between the cerebral cortex and the hypothalamus.
- It is involved in our behavioral and emotional responses, such as feeding,



reproduction, fighting, flight and fear. It is concerned with memory, olfaction and learning.

- **Limbic system is formed of two major components:**

- 1- **Cortical components (Limbic lobe):** Orbitofrontal cortex, cingulate gyrus, hippocampal formation, insular cortex & uncus.
- 2- **Subcortical components:** amygdala, olfactory bulb, hypothalamus & anterior & dorsomedial nuclei of thalamus & Septal nuclei.

- **Cingulate gyrus:**

- It lies on the medial surface of the cerebral cortex above the corpus callosum.

- **Hippocampal Formation:** consists of **hippocampus, dentate & para-hippocampal gyri.**

- **Hippocampus:** It is a sea horse-shaped curved elevation of gray matter that extends throughout the floor of the inferior horn of lateral ventricle.

- **Functions:**

- Converting recent memory into long-term memory.
- Spatial & Olfactory memories.
- **Its input:** from cingulate gyrus, olfactory-associated cortex & prefrontal association cortex.
- The **principal efferent** pathway is the **fornix**, which terminates mainly on **mammillary bodies & septal nuclei.**
- Damage to hippocampus leads to **retrograde amnesia.**

- **Amygdaloid nucleus:**

The amygdaloid nucleus is almond shaped.

It is situated **at the tip of the inferior horn of the lateral ventricle.** It is fused with the tip of tail of caudate nucleus.

Input: Olfactory-associated cortex & prefrontal association cortex.

